SIIMCS_BRD_V2

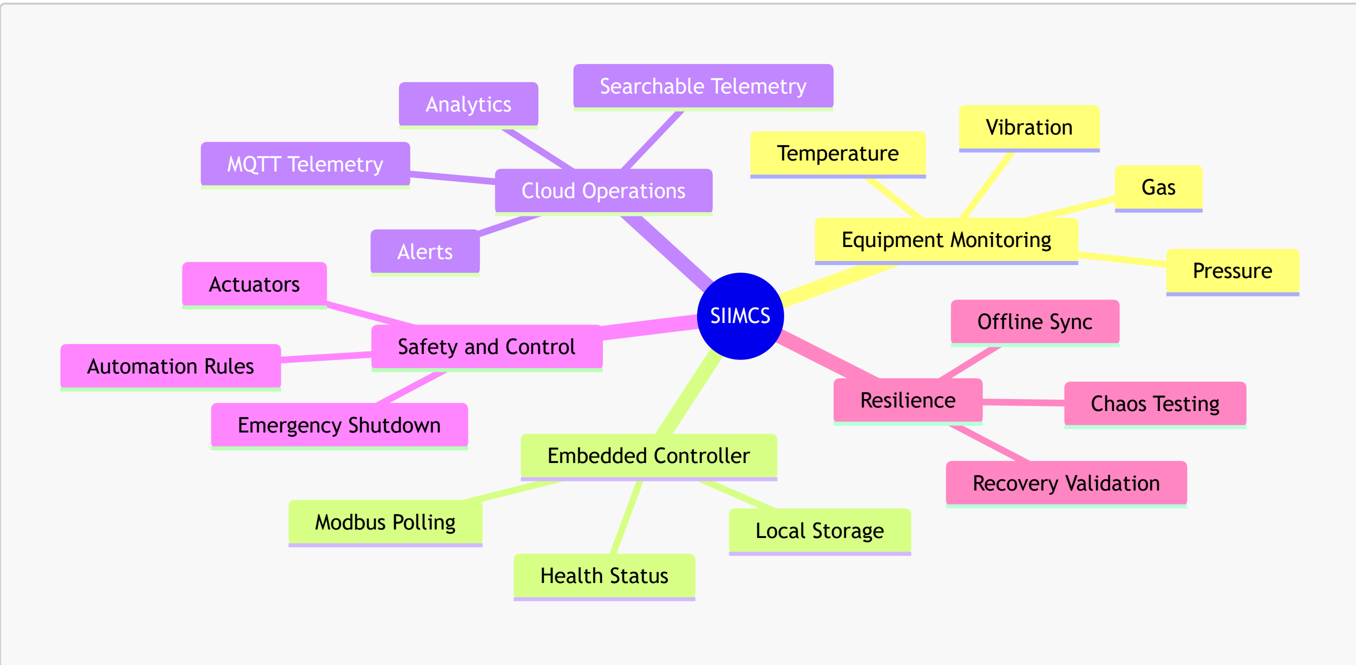
Smart Industrial IoT Monitoring & Control System Using Modbus Controller

1. Executive Summary

The Smart Industrial IoT Monitoring & Control System (SIIMCS) is a cloud-connected industrial monitoring and control solution for real-time equipment visibility, controlled automation, and maintenance planning.

This version extends the baseline BRD with three additional areas:

- MQTT telemetry assurance where MQTT is used for controller-to-cloud or edge-to-cloud communication.
- Controlled resilience validation using chaos testing.
- Searchable telemetry monitoring for faster incident review and operational analysis.



2. Business Objectives

Objective	Description
Real-Time Monitoring	Collect and present current operating data from industrial equipment.
Rule-Based Control	Trigger predefined actions when operating conditions cross approved limits.
Alerts and Response	Notify responsible teams when sensor values require attention.

Objective	Description
Integration	Share data with cloud platforms, dashboards, APIs, and enterprise systems.
Reliability	Maintain safe monitoring during sensor, controller, or connectivity issues.
Maintenance Support	Use historical readings, operating history, and equipment condition changes to plan maintenance and reduce unplanned downtime.
Operational Visibility	Make telemetry easy to search, review, and act on during live incidents and post-event analysis.
Resilience Assurance	Validate predictable system behavior during realistic failure conditions.

3. Business Scope

3.1 In Scope

- Controller-based polling of three configured sensors or field instruments.
- Collection of temperature, vibration, gas, pressure, or equivalent industrial readings.
- Cloud-based storage and monitoring.
- User-facing dashboards and operational interfaces.
- Alert and escalation workflows.
- Rule-based automation for safety and efficiency.
- REST API integration with third-party systems.
- Local data storage on the controller during internet outage.
- Synchronization of locally stored data after connectivity returns.
- Sensor and controller health visibility.
- MQTT-based telemetry checks where MQTT is part of the deployment.
- Searchable telemetry monitoring for operational review.
- Controlled chaos testing for resilience validation.

3.2 Out of Scope

- Manufacturing physical sensors.
- Full SCADA replacement.
- On-premises-only deployment in the initial phase.
- Custom AI model training beyond predefined industrial parameters.
- Full enterprise data lake design.
- Manual calibration workflow design.
- Custom hardware design for every industrial machine.

4. Stakeholders

Stakeholder	Responsibility
Client Organization	Owns business goals, priorities, and acceptance decisions.

Stakeholder	Responsibility
IT and Engineering Teams	Implement controller, cloud, interfaces, and integrations.
Operations Managers	Monitor equipment status, alerts, telemetry, and response actions.
Maintenance Teams	Use equipment condition data to plan service activity.
Safety Teams	Define critical limits and emergency actions.
Site Reliability and Support Teams	Validate resilience, recovery, and incident response readiness.
Third-Party Integrators	Connect SIIMCS with external enterprise systems.
QA and Validation Teams	Validate workflows, alerts, controls, and reliability scenarios.

---

5. System Overview

SIIMCS consists of:

- Industrial equipment being monitored.
- Sensors or field instruments connected to equipment.
- Embedded controller that reads sensor data using Modbus.
- Cloud services for storage, analytics, alerts, and reporting.
- User-facing dashboards and applications.
- REST APIs for enterprise integration.
- Optional MQTT telemetry path for cloud or edge communication.
- Searchable telemetry workspace for operations review.
- Actuators for approved automation and safety actions.

The controller acts as the Modbus client. Connected devices expose readings through their Modbus interface. Cloud-facing communication may use MQTT, HTTP, or both, depending on deployment design.

---

6. Functional Requirements

6.1 Sensor Data Collection and Monitoring

ID	Requirement
BR-SEN-001	The controller shall collect data from three configured sensors or field instruments.
BR-SEN-002	The controller shall poll each configured device at a defined interval.
BR-SEN-003	The system shall display the latest available readings to users.
BR-SEN-004	The system shall maintain current health status for each sensor and for the controller.
BR-SEN-005	Polling frequency shall be configurable within approved operating limits.

6.2 Supported Device Types

The initial release shall support three active sensor positions per controller deployment.

Device Type	Business Purpose
Temperature Sensor	Monitor heat levels in industrial machinery.
Vibration Sensor	Detect abnormal machine behavior and wear conditions.
Gas Sensor	Detect hazardous gas conditions.
Pressure Sensor	Monitor process pressure in pipes, tanks, or machines.

6.3 Communication and Polling

ID	Requirement
BR-COM-001	The controller shall initiate every Modbus transaction.
BR-COM-002	The controller shall continuously poll Sensor-1, Sensor-2, and Sensor-3 at configurable intervals.
BR-COM-003	Polling shall be deterministic and shall not allow one sensor to block the others.
BR-COM-004	The controller shall continue polling healthy sensors even if one sensor fails.
BR-COM-005	Communication failures shall update sensor health and availability status.

6.4 Startup and Configuration

ID	Requirement
BR-CFG-001	The controller shall validate configuration during startup.
BR-CFG-002	Duplicate addresses, invalid ranges, unsupported functions, or malformed configuration shall be rejected.
BR-CFG-003	Polling shall start only after configuration validation is successful.
BR-CFG-004	Authorized users shall be able to update thresholds and polling settings.
BR-CFG-005	If a runtime change fails validation, the last valid configuration shall remain active.

6.5 Request and Response Validation

ID	Requirement
BR-VAL-001	The controller shall use a sensor reading only when it matches a valid controller request.
BR-VAL-002	The controller shall validate device identity before accepting data.
BR-VAL-003	Invalid, delayed, duplicate, or mismatched responses shall not update business views or trigger actions.
BR-VAL-004	Communication errors shall update data quality and sensor health.

6.6 Offline Storage and Recovery

ID	Requirement
BR-OFF-001	If internet connectivity is lost, the controller shall store collected data locally.
BR-OFF-002	When connectivity returns, the controller shall synchronize stored data with the cloud.
BR-OFF-003	The local retention window shall be configurable, such as up to 48 hours.
BR-OFF-004	The application shall show cloud synchronization status.

6.7 Alerts and Notifications

ID	Requirement
BR-ALT-001	Users shall be able to configure warning and critical thresholds.
BR-ALT-002	Alerts shall be available through operational interfaces and configurable notification channels.
BR-ALT-003	Event-driven alerts shall be supported for abnormal equipment conditions.
BR-ALT-004	Escalation shall occur if an alert is not acknowledged within configured time limits.
BR-ALT-005	Alerts shall include equipment, sensor type, value, severity, time, and recommended action.

6.8 Rule-Based Automation

The system shall support approved automation rules based on equipment condition.

Condition	Safety Level	Expected Action
Temperature exceeds threshold	Medium	Increase fan speed.
Vibration exceeds safety limit	Medium	Reduce motor speed.
Gas leakage detected	High	Trigger emergency shutdown.
Temperature warning and vibration high	Medium	Trigger warning and reduce machine speed by 20%.
Pressure low and temperature critical	High	Emergency shutdown and notify operators.

6.9 Sensor and Actuator Data Sheet

Type	Normal Range	Min Threshold	Max Threshold	Critical Level
Temperature Sensor	10-50°C	5-10°C	50-70°C	>70°C
Vibration Sensor	1-5 mm/s	0-1 mm/s	5-8 mm/s	>8 mm/s
Gas Sensor	10-12 psi	5-10 psi	12-13 psi	>13 psi
Pressure Sensor	30-50 psi	10-30 psi	50-60 psi	<10 psi or >60 psi
Fan Actuator	Variable Speed	Low Speed	High Speed	Max Speed
Motor Controller	0-100% Speed	10% Speed	90% Speed	Full Stop
Valve Controller	Open/Close	10% Open	90% Open	Emergency Shut
Shutdown Relay	Active/Inactive	Warning Level	Critical Level	Full Shutdown

---

6.10 Health and Data Status

ID	Requirement
BR-HLT-001	Sensor health shall be shown as Online, Degraded, Offline, or Protocol Error.
BR-HLT-002	One failed sensor shall not stop monitoring of the remaining sensors.
BR-DQ-001	Data shall be identified as Good, Stale, Invalid, or Unavailable.
BR-DQ-002	Latest good value shall be retained during temporary failure.
BR-DQ-003	Stale data shall be clearly identified in dashboards and APIs.

6.11 User Experience and Integration

ID	Requirement
BR-APP-001	The system shall display current readings, health, alerts, and recent status.
BR-APP-002	The system shall expose sensor data and status through REST APIs.

ID	Requirement
BR-APP-003	Third-party systems shall be able to integrate through approved secure interfaces.

6.12 MQTT Telemetry Assurance

This section applies when MQTT is used between the controller, edge services, or cloud services for telemetry transport.

ID	Requirement
BR-MQTT-001	The system shall verify broker connectivity and session health for MQTT-based telemetry paths.
BR-MQTT-002	The system shall validate topic design, delivery expectations, and message quality level for critical telemetry.
BR-MQTT-003	The system shall detect disconnect, reconnect, session expiry, and backlog conditions that may affect telemetry visibility.
BR-MQTT-004	The system shall confirm that telemetry published during intermittent connectivity is replayed, marked as missing, or handled by approved business policy.
BR-MQTT-005	The system shall expose MQTT communication state in operational interfaces when MQTT is part of the deployment.

Recommended business checks:

- Broker reachable from approved network paths.
- Messages arrive on expected topics.
- Delivery quality matches business criticality.
- Reconnect behavior restores expected telemetry flow.
- Missed or delayed telemetry is visible to operators.

6.13 Searchable Telemetry Monitoring

The system should support a searchable telemetry workspace for operations review where needed, using search and analytics platforms such as tools like open-search.

ID	Requirement
BR-OBS-001	The system should make telemetry searchable by site, controller, asset, sensor, severity, and time range.
BR-OBS-002	Operations teams should be able to query warning, critical, and shutdown events during incident review.
BR-OBS-003	Telemetry views should support live monitoring and historical investigation.

ID	Requirement
BR-OBS-004	The searchable telemetry solution should support MQTT-fed telemetry where MQTT is part of the platform design.

Example use case:

If a pressure alert occurs at 09:05, operations should be able to search all related controller events, sensor values, acknowledgements, and recovery status for that equipment over the surrounding time window.

6.14 Security

ID	Requirement
BR-SEC-001	Only authorized users shall configure sensors, thresholds, and automation rules.
BR-SEC-002	Access to APIs and control actions shall be authenticated and authorized.
BR-SEC-003	Network access to controller and cloud services shall follow enterprise security policy.
BR-SEC-004	Safety-critical actions shall require approved business rules.
BR-SEC-005	MQTT and cloud telemetry paths shall follow approved transport security policy where used.

---

7. Non-Functional Requirements

Area	Requirement
Reliability	Monitoring shall continue safely when one sensor or one communication path fails.
Availability	Latest good data shall remain available during temporary disruptions.
Performance	Polling, alerting, synchronization, and cloud telemetry shall meet agreed business service levels.
Scalability	The design shall support future expansion to more assets and devices.
Maintainability	Thresholds, rules, and polling settings shall be configurable.
Security	Access to data and control actions shall be protected.
Embedded Safety	The controller shall operate within approved CPU, memory, storage, and watchdog limits.
Recoverability	The system shall recover predictably from controller restarts, reconnects, and temporary cloud failures.

---

8. Resilience and Chaos Validation

Chaos testing is useful for SIIMCS when it is controlled, targeted, and tied to business outcomes.

Required scenarios:

- Sensor timeout during active monitoring.
- One sensor returning invalid data while other sensors remain healthy.
- Controller restart during telemetry flow.
- Internet loss during data collection.
- MQTT broker disconnect or reconnect where MQTT is used.
- Cloud ingestion delay or temporary outage.
- Alert backlog or delayed acknowledgement.
- Local storage approaching retention limit.

ID	Requirement
BR-CHAOS-001	Chaos tests shall validate that monitoring remains understandable and recoverable under adverse conditions.
BR-CHAOS-002	Chaos tests shall be executed first in pre-production or controlled environments that resemble production.
BR-CHAOS-003	Each experiment shall define expected steady-state business behavior before faults are introduced.
BR-CHAOS-004	Each experiment shall record impact on data freshness, alerts, automation, and recovery time.
BR-CHAOS-005	Fault injection shall never bypass safety approval for production-like environments.

9. Acceptance Criteria

ID	Acceptance Criteria
AC-001	The controller polls three configured sensors successfully.
AC-002	Current readings, health, and alerts are visible to users.
AC-003	One sensor failure does not stop the remaining sensors from being monitored.
AC-004	Warning and critical alerts trigger correctly.
AC-005	Approved automation rules trigger the correct control actions.
AC-006	Data is stored locally during internet outage and synchronized after recovery.
AC-007	Runtime configuration updates are validated before activation.
AC-008	Security and access controls are enforced.
AC-009	Long-duration monitoring runs without controller crash or data loss.

ID	Acceptance Criteria
AC-010	MQTT communication checks pass for deployments that use MQTT.
AC-011	Searchable telemetry supports incident review for critical events.
AC-012	Controlled chaos scenarios confirm recovery behavior and operational visibility.

---

10. Definition of Done

The feature is complete when:

- Three configured sensors are monitored successfully.
- Dashboards, alerts, APIs, and automation rules are working.
- Offline storage and data synchronization are verified.
- MQTT communication checks are verified where MQTT is used.
- Searchable telemetry flows support operational review where enabled.
- Sensor, controller, cloud, and network disruption scenarios are tested.
- Controlled chaos scenarios are executed and reviewed.
- Security and access controls are verified.
- Stakeholders approve the business requirements and acceptance criteria.